

DRY-RUN GUIDE | prefix_sum.py
Source: Python\4th_belt\prefix_sum.py

CORE BEHAVIOR

Builds inclusive cumulative sums from left to right.

REPRESENTATIVE INPUT

[2,1,3,4].

STEP TRACE

Running totals: 2, 3, 6, 10.

EXPECTED OUTPUT / RETURN

[2,3,6,10]

ACTUAL SOURCE EXCERPT

Verbatim excerpt from Python\4th_belt\prefix_sum.py; line numbers refer to the original file. No source was modified.

```
001 | # Question: Compute the prefix sum of each element in the array.
```

```
002 | # Add your solution here.
```

```
003 |
```

```
004 | a = [3, 4, 1, 7, 9, 1]
```

```
005 |
```

```
006 | res = []
```

```
007 | cur_s= 0 # current sum
```

```
008 |
```

```
009 | for num in a:
```

```
010 |     cur_s += num
```

```
011 |     res.append(cur_s)
```

```
012 | print(res)
```

```
013 |
```

```
014 | # or
```

```
015 |
```

```
016 | import numpy as np
```

```
017 | a = [3, 4, 1, 7, 9, 1]
```

```
018 |
```

```
019 | res = np.cumsum(a)
```

```
020 | print(res)
```

```
021 |
```

```
022 | # or
```

```
023 |
```

```
024 | import itertools
```

```
025 | a = [3, 4, 1, 7, 9, 1]
```

```
026 |
```

```
027 | res = list(itertools.accumulate(a))
```

```
028 | print(res)
```

```
029 |
```

```
030 | # or
```

```
031 |
```

```
032 | a = [3, 4, 1, 7, 9, 1]
```

```
033 | res = [sum(a[:i+1]) for i in range(len(a))]
```

```
034 |
```

```
035 | print(res)
```

DRY-RUN NOTE

Values are representative. For function-only modules the guide uses an illustrative direct call and explains its return; quirks/errors describe actual behavior.